# System design notes

Loops, faucets, drains, feedback, and the failure modes I check first.

Sector: Systems design, all sectors

Status: concept study, written and designed by Karthik Bindumon, 2026. Numbers are targets or modelled assumptions and are labelled as such.

Method: The Realisation Loop, v1 (see 00-realisation-loop.pdf).

Every game, gamified product and learning programme I build is a set of loops. These notes are the method, with the one-currency toy model on the site written out, and the failure modes I check for before a single screen is designed.

## 1 The one-page loop

Name four things: the currency (what the player accumulates), the faucet (how it enters), the drain (how it leaves) and the feedback (what the player sees at the moment of a decision). If there is no drain, there is no economy, only a pile. If feedback arrives later than the decision, the loop teaches nothing.

## 2 Failure modes, checked in this order

| Mode | Symptom | Fix |
|---|---|---|
| Inflation | Wallet hits the cap; prices feel free; the store dies; nobody spends. | Add or scale a drain; introduce a second currency with a real trade-off. |
| Starvation | A wall by a predictable day; churn or pay, and most churn. | Raise the faucet late in session, not early; never at the wall itself. |
| Drift | Loop mastered; fun stops; nothing new to decide. | Change the decision, not the reward; a new constraint beats a new prize. |
| Dependence | System only works while the designer tunes it. | Expose dials to live-ops; write the tuning rules down. |

## 3 The toy model

Balance(t+1) = Balance(t) + faucet - drain, starting purse 200, wallet cap 1,000, thirty days. Starvation is a balance at or below zero; inflation is a balance at the cap. Tension held is a faucet-to-drain ratio between 0.87 and 1.15, the range where every purchase is a decision. The real model has four currencies, a session-length curve and a live-ops sheet; the toy shows the argument.

## 4 The same discipline in the learning cases

| Case | Currency | Faucet | Drain | Feedback |
|---|---|---|---|---|
| Threshold | Clean entries | Each rub at the tile | Twelve raise the flag; the count resets | Tile colour, click, flag |
| Kata | Stamps | Each kata passed | Ri: teach it away to the next hire | Stamp on the card |

| Case | Currency | Faucet | Drain | Feedback |
|---|---|---|---|---|
| Rep | Beads | Each rep | The marker comes home; the set resets | The knot |
| Salary Day | Rupees | Payday | Every choice | The reveal at month 36 |

## 5 Ethics of loops

Variable reward is a tool with a narrow use: seeking behaviour in low-stakes play. It is out of scope for health, money, safety and anything a person must do while hurt or tired. A loop the player cannot see is a trap, not a system.